Project 1: Exploratory Data Analysis and Simple Regression

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Overview

This report presents an exploratory data analysis (EDA) and simple (univariate) linear regression study on three datasets from the UCI Machine Learning Repository: Auto MPG, Concrete Compressive Strength, and Airfoil Self-Noise. For each dataset we (1) describe preprocessing steps, (2) report statistical summaries for every feature, (3) compute a correlation matrix and heatmap, and (4) fit a simple linear regression model, $y = b_0 + b_1 x$, for the two predictor features most strongly correlated with the target, using both **ScalaTion** and Python's **Statsmodels**.

1 Auto MPG Dataset

1.1 Description

The Auto MPG dataset relates a car's fuel economy (miles per gallon, mpg) to seven engine and body characteristics. After cleaning, it contains 392 rows and 8 columns.

1.2 Preprocessing

The raw `auto-mpg.data` file is whitespace-delimited and contains 398 rows and 9 columns, the last of which (`car name`) is a unique string identifier for each car and is not useful as a regression feature. Six rows have a missing (`?`) value for `horsepower`; these were dropped during the conversion to CSV, along with the `car name` column, leaving 392 rows and 8 numeric columns (`mpg`, `cylinders`, `displacement`, `horsepower`, `weight`, `acceleration`, `model_year`, `origin`) exactly matching the expected shape. No further outlier removal was performed: an IQR-rule check flags 10 `horsepower` values and 11 `acceleration` values as statistical outliers, but these correspond to genuinely high-performance or economy vehicles rather than data errors, so they were kept.

1.3 Statistical Summary

Feature	count	mean	std	min	25%	50%	75%	max
mpg	392.000	23.446	7.805	9.000	17.000	22.750	29.000	46.600
cylinders	392.000	5.472	1.706	3.000	4.000	4.000	8.000	8.000
displacement	392.000	194.412	104.644	68.000	105.000	151.000	275.750	455.000
horsepower	392.000	104.469	38.491	46.000	75.000	93.500	126.000	230.000
weight	392.000	2977.584	849.403	1613.000	2225.250	2803.500	3614.750	5140.000
acceleration	392.000	15.541	2.759	8.000	13.775	15.500	17.025	24.800
model_year	392.000	75.980	3.684	70.000	73.000	76.000	79.000	82.000
origin	392.000	1.577	0.806	1.000	1.000	1.000	2.000	3.000

1.4 Correlation Analysis

Feature	Correlation with target
weight	-0.8322
displacement	-0.8051
horsepower	-0.7784
cylinders	-0.7776
model_year	0.5805
origin	0.5652
acceleration	0.4233

Table 1 gives the complete feature-by-feature correlation matrix, and Figure 1 shows the corresponding heatmap.

Table 1: Full correlation matrix, Auto MPG dataset.

Feature	mpg	C	D	H	W	A	MY	O
mpg	1.00	-0.78	-0.81	-0.78	-0.83	0.42	0.58	0.57
cylinders	-0.78	1.00	0.95	0.84	0.90	-0.50	-0.35	-0.57
displacement	-0.81	0.95	1.00	0.90	0.93	-0.54	-0.37	-0.61
horsepower	-0.78	0.84	0.90	1.00	0.86	-0.69	-0.42	-0.46
weight	-0.83	0.90	0.93	0.86	1.00	-0.42	-0.31	-0.59
acceleration	0.42	-0.50	-0.54	-0.69	-0.42	1.00	0.29	0.21
model_year	0.58	-0.35	-0.37	-0.42	-0.31	0.29	1.00	0.18
origin	0.57	-0.57	-0.61	-0.46	-0.59	0.21	0.18	1.00

C=cylinders; D=displacement; H=horsepower; W=weight; A=acceleration; MY=model_year; O=origin

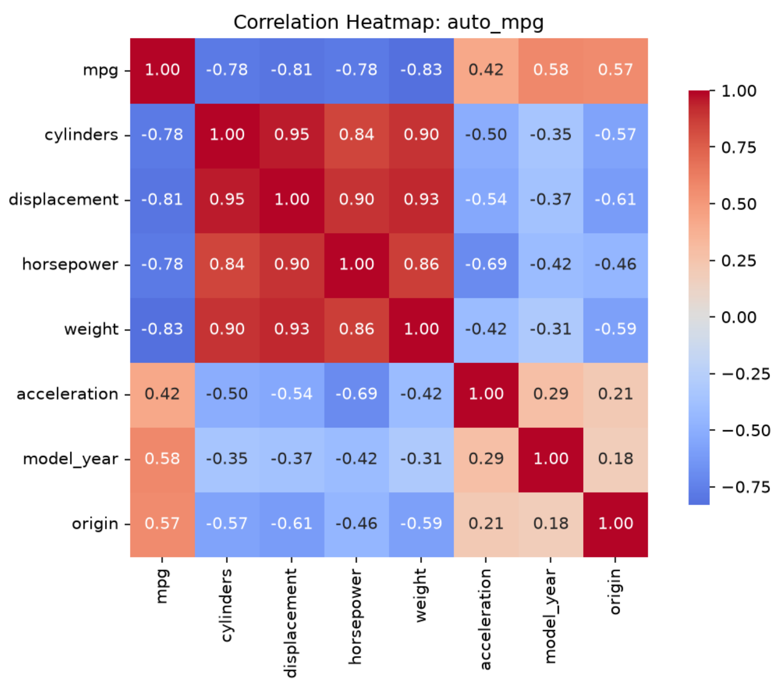


Figure 1: Correlation heatmap for the Auto MPG dataset.

1.5 Top Two Predictor Variables

Based on the correlation analysis above, the two predictors most strongly correlated (by magnitude) with `mpg` are `weight` ($r = -0.83$) and `displacement` ($r = -0.81$), ahead of `horsepower` ($r = -0.78$) and `cylinders` ($r = -0.78$). These two features are used consistently in both the ScalaTion and Statsmodels simple regressions below.

1.6 Simple Regression Models

Feature: weight

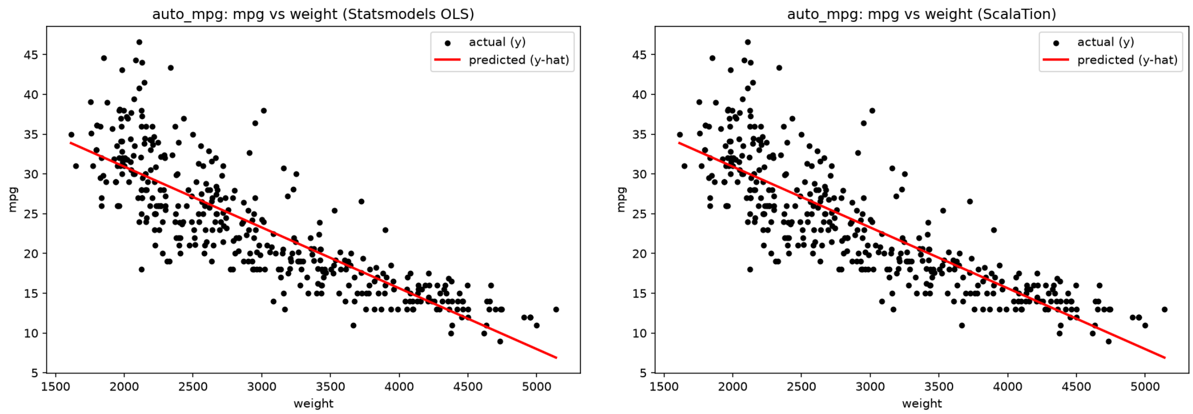
ScalaTion fit report:

```
REPORT
    ----------------------------------------------------------------------------
    modelName mn = SimpleRegression_weight
    ----------------------------------------------------------------------------
    hparameter hp = null
    ----------------------------------------------------------------------------
    features fn = Array(one, weight)
    ----------------------------------------------------------------------------
    parameter b = VectorD(46.2165, -0.00764734)
    ----------------------------------------------------------------------------
    fitMap qof = LinkedHashMap(rSq -> 0.692630, rSqBar -> 0.691842, sst -> 23818.993469, sse -> 7321.233706,
        sde -> 4.327168, rse -> 4.332712, mse0 -> 18.676617, rmse -> 4.321645, mae -> 3.278702, smape ->
        14.140828, bmae -> 29.224578, m -> 392.000000, dfr -> 1.000000, df -> 390.000000, fStat ->
        878.830886, aic -> -1125.969274, bic -> -1118.026751, mape -> 14.413211, mase -> 1.175904, smapeC ->
        14.151032, picp -> -1.000000, pinc -> -1.000000, ace -> -1.000000, pinaw -> -1.000000, mis ->
        -1.000000, wis -> -1.000000)
    ----------------------------------------------------------------------------

fname = Array(one, weight)
SUMMARY
    Parameters/Coefficients:
    Var Estimate Std. Error  t value  Pr(>|t|)  VIF
--------------------------------------------------------------------------------
    x0 46.216525 0.798672 57.866681 0.000000 NA
    x1 -0.007647 0.000258 -29.645082 0.000000 1.000000

    Residual standard error: 4.332712 on 390.0 degrees of freedom
    Multiple R-squared: 0.692630, Adjusted R-squared: 0.691842
    F-statistic: 878.8308864132066 on 1.0 and 390.0 DF, p-value: -0.0
--------------------------------------------------------------------------------
```

Statsmodels fit report:

```
                          OLS Regression Results
==============================================================================
Dep. Variable: y R-squared: 0.693
Model: OLS Adj. R-squared: 0.692
Method: Least Squares F-statistic: 878.8
Date: Wed, 02 Sep 2026 Prob (F-statistic): 6.02e-102
Time: 16:40:19 Log-Likelihood: -1130.0
No. Observations: 392 AIC: 2264.
Df Residuals: 390 BIC: 2272.
Df Model: 1
Covariance Type: nonrobust
==============================================================================
               coef std err t P>|t| [0.025 0.975]
------------------------------------------------------------------------------
const 46.2165 0.799 57.867 0.000 44.646 47.787
x1 -0.0076 0.000 -29.645 0.000 -0.008 -0.007
==============================================================================
Omnibus: 41.682 Durbin-Watson: 0.808
Prob(Omnibus): 0.000 Jarque-Bera (JB): 60.039
Skew: 0.727 Prob(JB): 9.18e-14
Kurtosis: 4.251 Cond. No. 1.13e+04
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 1.13e+04. This might indicate that there are
strong multicollinearity or other numerical problems.
```

Summary statement: ScalaTion and Statsmodels landed on the same fitted line: $\hat{y} = 46.2165 - 0.0076\,x$. Every extra pound of vehicle weight costs about 0.0076 mpg, and the fit is reasonably strong for a single predictor, explaining 69.3% of the variance in mpg ($R^2 = 0.6926$, adj. $R^2 = 0.6918$). The slope is highly significant ($F = 878.83$, $p < 0.001$).



(a) Statsmodels

(b) ScalaTion

Figure 2: mpg vs. weight: actual (black) and predicted (red) values.

Feature: displacement

ScalaTion fit report:

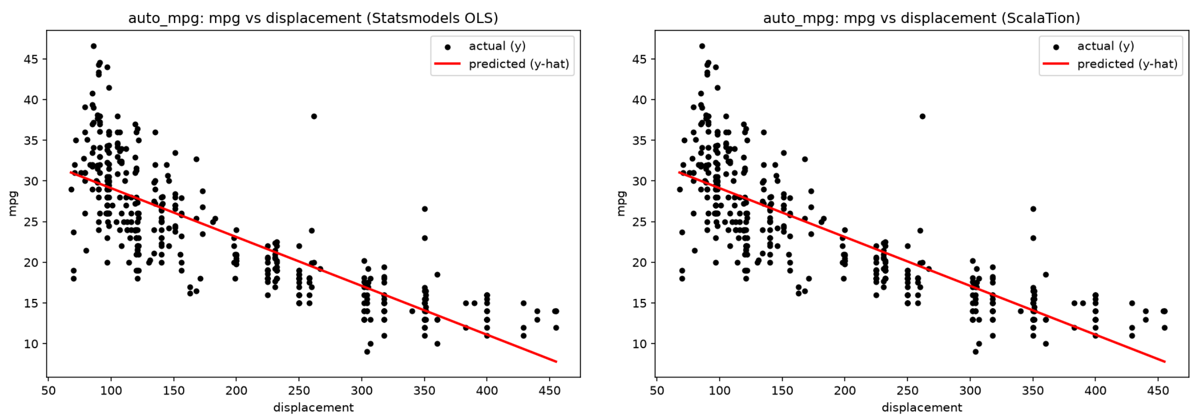
```
REPORT
    ----------------------------------------------------------------------------
    modelName mn = SimpleRegression_displacement
    ----------------------------------------------------------------------------
    hparameter hp = null
    ----------------------------------------------------------------------------
    features fn = Array(one, displacement)
    ----------------------------------------------------------------------------
    parameter b = VectorD(35.1206, -0.0600514)
    ----------------------------------------------------------------------------
    fitMap qof = LinkedHashMap(rSq -> 0.648229, rSqBar -> 0.647327, sst -> 23818.993469, sse -> 8378.821617,
         sde -> 4.629170, rse -> 4.635101, mse0 -> 21.374545, rmse -> 4.623261, mae -> 3.506273, smape ->
         15.257579, bmae -> 30.616313, m -> 392.000000, dfr -> 1.000000, df -> 390.000000, fStat ->
         718.677076, aic -> -1152.415247, bic -> -1144.472723, mape -> 15.556335, mase -> 1.231903, smapeC ->
         15.267783, picp -> -1.000000, pinc -> -1.000000, ace -> -1.000000, pinaw -> -1.000000, mis ->
         -1.000000, wis -> -1.000000)
    ----------------------------------------------------------------------------

fname = Array(one, displacement)
SUMMARY
    Parameters/Coefficients:
    Var Estimate Std. Error  t value  Pr(>|t|)  VIF
--------------------------------------------------------------------------------
    x0 35.120636 0.494428 71.032840 0.000000 NA
    x1 -0.060051 0.002240 -26.808153 0.000000 1.000000

    Residual standard error: 4.635101 on 390.0 degrees of freedom
    Multiple R-squared: 0.648229, Adjusted R-squared: 0.647327
    F-statistic: 718.6770763503415 on 1.0 and 390.0 DF, p-value: -0.0
--------------------------------------------------------------------------------
```

Statsmodels fit report:

```
                         OLS Regression Results
==============================================================================
Dep. Variable: y R-squared: 0.648
Model: OLS Adj. R-squared: 0.647
Method: Least Squares F-statistic: 718.7
Date: Wed, 02 Sep 2026 Prob (F-statistic): 1.66e-90
Time: 16:40:20 Log-Likelihood: -1156.4
No. Observations: 392 AIC: 2317.
```

```
Df Residuals: 390 BIC: 2325.
Df Model: 1
Covariance Type: nonrobust
==============================================================================
               coef std err t P>|t| [0.025 0.975]
------------------------------------------------------------------------------
const 35.1206 0.494 71.033 0.000 34.149 36.093
x1 -0.0601 0.002 -26.808 0.000 -0.064 -0.056
==============================================================================
Omnibus: 41.308 Durbin-Watson: 0.926
Prob(Omnibus): 0.000 Jarque-Bera (JB): 61.139
Skew: 0.709 Prob(JB): 5.30e-14
Kurtosis: 4.317 Cond. No. 466.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

Summary statement: The fitted line, $\hat{y} = 35.1206 - 0.0601\,x$, is identical between the two tools. Each additional cubic inch of displacement costs about 0.06 mpg, and the model explains 64.8% of the variance ($R^2 = 0.6482$, adj. $R^2 = 0.6473$), with a highly significant slope ($F = 718.68$, $p < 0.001$).



(a) Statsmodels

(b) ScalaTion

Figure 3: mpg vs. displacement: actual (black) and predicted (red) values.

1.7 Discussion

Both predictors show a strong, statistically significant negative relationship with fuel economy: heavier cars with larger engines get fewer miles per gallon, which lines up with basic physics, since moving more mass takes more fuel. `weight` is the better single predictor ($R^2 \approx 0.69$ versus 0.65 for `displacement`), though the two are correlated with each other at $r = 0.93$ and so are largely capturing the same thing: overall vehicle size. ScalaTion and Statsmodels produced identical coefficients and R^2 values, which is expected for the same least-squares problem and confirms both were set up correctly. Neither model explains more than 70% of the variance in `mpg` on its own, so a multiple regression combining several of these features, or a nonlinear model, would likely fit better.

2 Concrete Compressive Strength Dataset

2.1 Description

This dataset relates the compressive strength of concrete (MPa) to the proportions of its ingredients and its curing age. It contains 1030 rows and 9 columns (8 quantitative ingredient/age features plus the strength target).

2.2 Preprocessing

The raw data ships as an Excel spreadsheet (`Concrete_Data.xls`) with verbose column headers. It was loaded with pandas, renamed to concise snake_case names (`cement`, `blast_furnace_slag`, `fly_ash`, `water`, `superplasticizer`, `coarse_aggregate`, `fine_aggregate`, `age`, `concrete_compressive_strength`), and written out as a plain CSV. The dataset has no missing values and no string columns, so no rows were dropped; its cleaned shape (1030, 9) matches the documented dataset size exactly. An IQR-rule check flags outliers in several columns, most prominently `age` (59 flagged values, since age ranges from 1 to 365 days and is heavily right-skewed). These were kept, since they represent valid experimental conditions rather than data-entry errors.

2.3 Statistical Summary

Feature	count	mean	std	min	25%	50%	75%	max
cement	1030.000	281.166	104.507	102.000	192.375	272.900	350.000	540.000
blast_furnace_slag	1030.000	73.895	86.279	0.000	0.000	22.000	142.950	359.400
fly_ash	1030.000	54.187	63.996	0.000	0.000	0.000	118.270	200.100
water	1030.000	181.566	21.356	121.750	164.900	185.000	192.000	247.000
superplasticizer	1030.000	6.203	5.973	0.000	0.000	6.350	10.160	32.200
coarse_aggregate	1030.000	972.919	77.754	801.000	932.000	968.000	1029.400	1145.000
fine_aggregate	1030.000	773.579	80.175	594.000	730.950	779.510	824.000	992.600
age	1030.000	45.662	63.170	1.000	7.000	28.000	56.000	365.000
concrete_compressive_strength	1030.000	35.818	16.706	2.332	23.707	34.443	46.136	82.599

2.4 Correlation Analysis

Feature	Correlation with target
cement	0.4978
superplasticizer	0.3661
age	0.3289
water	-0.2896
fine_aggregate	-0.1672
coarse_aggregate	-0.1649
blast_furnace_slag	0.1348
fly_ash	-0.1058

Table 2 gives the complete feature-by-feature correlation matrix, and Figure 4 shows the corresponding heatmap.

Table 2: Full correlation matrix, Concrete Compressive Strength dataset.

Feature	C	BFS	FA	W	S	CA	FA2	age	CCS
cement	1.00	-0.28	-0.40	-0.08	0.09	-0.11	-0.22	0.08	0.50
blast_furnace_slag	-0.28	1.00	-0.32	0.11	0.04	-0.28	-0.28	-0.04	0.13
fly_ash	-0.40	-0.32	1.00	-0.26	0.38	-0.01	0.08	-0.15	-0.11
water	-0.08	0.11	-0.26	1.00	-0.66	-0.18	-0.45	0.28	-0.29
superplasticizer	0.09	0.04	0.38	-0.66	1.00	-0.27	0.22	-0.19	0.37
coarse_aggregate	-0.11	-0.28	-0.01	-0.18	-0.27	1.00	-0.18	-0.00	-0.16
fine_aggregate	-0.22	-0.28	0.08	-0.45	0.22	-0.18	1.00	-0.16	-0.17
age	0.08	-0.04	-0.15	0.28	-0.19	-0.00	-0.16	1.00	0.33
concrete_compressive_strength	0.50	0.13	-0.11	-0.29	0.37	-0.16	-0.17	0.33	1.00

C=cement; BFS=blast_furnace_slag; FA=fly_ash; W=water; S=superplasticizer; CA=coarse_aggregate; FA2=fine_aggregate; CCS=concrete_compressive_strength

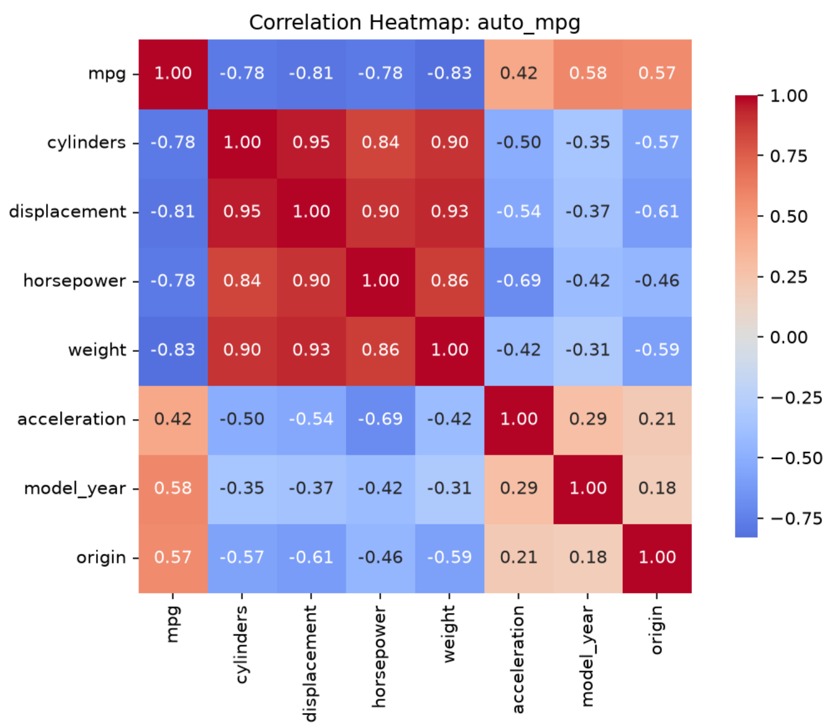


Figure 4: Correlation heatmap for the Concrete Compressive Strength dataset.

2.5 Top Two Predictor Variables

Based on the correlation analysis above, the two predictors most strongly correlated with compressive strength are `cement` ($r = 0.50$) and `superplasticizer` ($r = 0.37$), ahead of `age` ($r = 0.33$) and `water` ($r = -0.29$). These two features are used consistently in both the ScalaTion and Statsmodels simple regressions below.

2.6 Simple Regression Models

Feature: cement

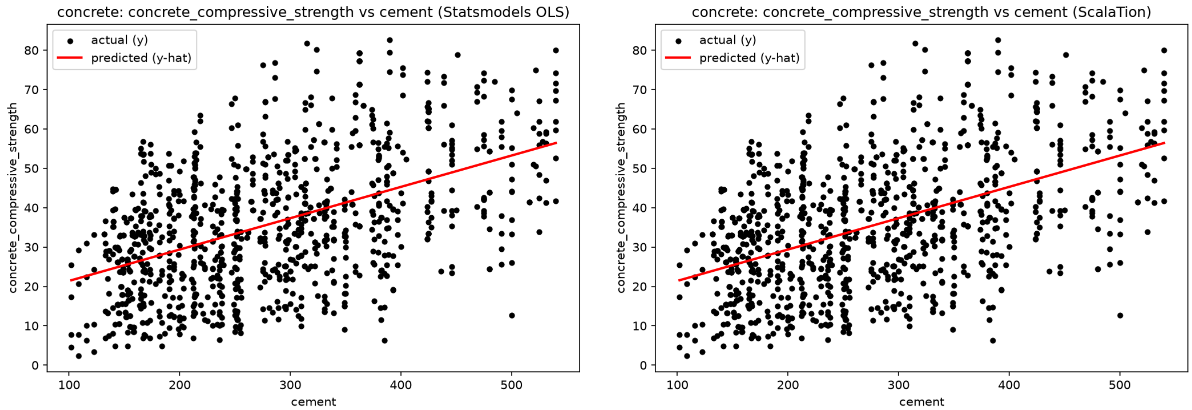
ScalaTion fit report:

```
REPORT
    ----------------------------------------------------------------------------
    modelName mn = SimpleRegression_cement
    ----------------------------------------------------------------------------
    hparameter hp = null
    ----------------------------------------------------------------------------
    features fn = Array(one, cement)
    ----------------------------------------------------------------------------
    parameter b = VectorD(13.4428, 0.0795796)
    ----------------------------------------------------------------------------
    fitMap qof = LinkedHashMap(rSq -> 0.247837, rSqBar -> 0.247106, sst -> 287173.028478, sse ->
        216000.806188, sde -> 14.488386, rse -> 14.495431, mse0 -> 209.709521, rmse -> 14.481351, mae ->
        11.851871, smape -> 36.311902, bmae -> 49.246705, m -> 1030.000000, dfr -> 1.000000, df ->
        1028.000000, fStat -> 338.725794, aic -> -4210.554208, bic -> -4200.679580, mape -> 49.910284, mase
        -> 1.095895, smapeC -> 36.315785, picp -> -1.000000, pinc -> -1.000000, ace -> -1.000000, pinaw ->
        -1.000000, mis -> -1.000000, wis -> -1.000000)
    ----------------------------------------------------------------------------

fname = Array(one, cement)
SUMMARY
    Parameters/Coefficients:
    Var Estimate Std. Error  t value  Pr(>|t|)  VIF
--------------------------------------------------------------------------------
    x0 13.442795 1.296925 10.365129 0.000000 NA
    x1 0.079580 0.004324 18.404505 0.000000 1.000000

    Residual standard error: 14.495431 on 1028.0 degrees of freedom
    Multiple R-squared: 0.247837, Adjusted R-squared: 0.247106
    F-statistic: 338.7257936916475 on 1.0 and 1028.0 DF, p-value: -0.0
--------------------------------------------------------------------------------
```

Statsmodels fit report:

```
                         OLS Regression Results
==============================================================================
Dep. Variable: y R-squared: 0.248
Model: OLS Adj. R-squared: 0.247
Method: Least Squares F-statistic: 338.7
Date: Wed, 02 Sep 2026 Prob (F-statistic): 1.32e-65
Time: 16:40:24 Log-Likelihood: -4214.6
No. Observations: 1030 AIC: 8433.
Df Residuals: 1028 BIC: 8443.
Df Model: 1
Covariance Type: nonrobust
==============================================================================
               coef std err t P>|t| [0.025 0.975]
------------------------------------------------------------------------------
const 13.4428 1.297 10.365 0.000 10.898 15.988
x1 0.0796 0.004 18.405 0.000 0.071 0.088
==============================================================================
Omnibus: 19.695 Durbin-Watson: 1.012
Prob(Omnibus): 0.000 Jarque-Bera (JB): 17.890
Skew: 0.271 Prob(JB): 0.000130
Kurtosis: 2.649 Cond. No. 861.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

Summary statement: ScalaTion and Statsmodels again agree exactly: $\hat{y} = 13.4428+0.0796\,x$. Each extra kg/m^3 of cement adds roughly 0.08 MPa of compressive strength, though the model only accounts for 24.8% of the variance ($R^2 = 0.2478$, adj. $R^2 = 0.2471$). The slope itself is still highly significant ($F = 338.73$, $p < 0.001$).



(a) Statsmodels

(b) ScalaTion

Figure 5: Compressive strength vs. cement content: actual (black) and predicted (red) values.

Feature: superplasticizer

ScalaTion fit report:

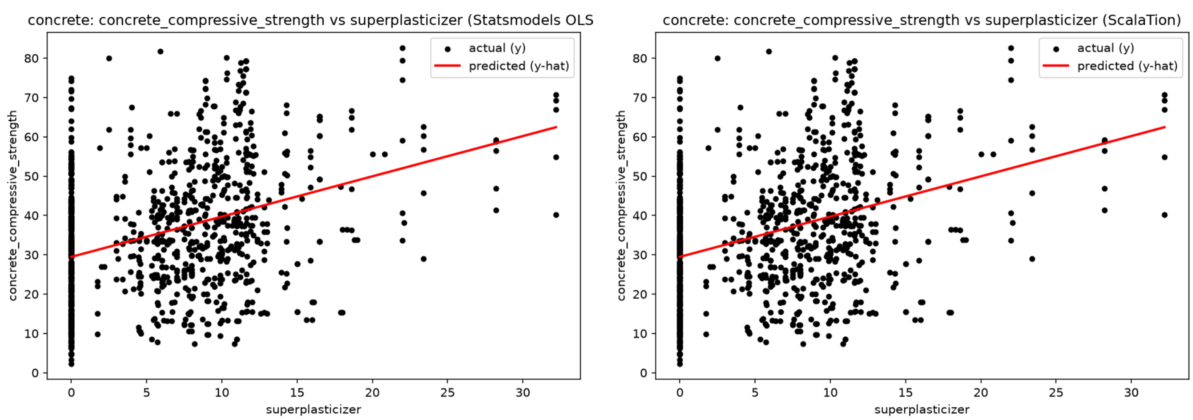
```
REPORT
    ----------------------------------------------------------------------------
    modelName mn = SimpleRegression_superplasticizer
    ----------------------------------------------------------------------------
    hparameter hp = null
    ----------------------------------------------------------------------------
    features fn = Array(one, superplasticizer)
    ----------------------------------------------------------------------------
    parameter b = VectorD(29.4668, 1.02385)
    ----------------------------------------------------------------------------
    fitMap qof = LinkedHashMap(rSq -> 0.134031, rSqBar -> 0.133189, sst -> 287173.028478, sse ->
         248682.971207, sde -> 15.545881, rse -> 15.553440, mse0 -> 241.439778, rmse -> 15.538333, mae ->
         12.615474, smape -> 38.158411, bmae -> 48.282956, m -> 1030.000000, dfr -> 1.000000, df ->
         1028.000000, fStat -> 159.109322, aic -> -4283.116028, bic -> -4273.241400, mape -> 54.595327, mase
         -> 1.074448, smapeC -> 38.162294, picp -> -1.000000, pinc -> -1.000000, ace -> -1.000000, pinaw ->
         -1.000000, mis -> -1.000000, wis -> -1.000000)
    ----------------------------------------------------------------------------

fname = Array(one, superplasticizer)
SUMMARY
    Parameters/Coefficients:
    Var Estimate Std. Error  t value  Pr(>|t|)  VIF
--------------------------------------------------------------------------------
    x0 29.466751 0.698839 42.165264 0.000000 NA
    x1 1.023855 0.081169 12.613854 0.000000 1.000000

    Residual standard error: 15.553440 on 1028.0 degrees of freedom
    Multiple R-squared: 0.134031, Adjusted R-squared: 0.133189
    F-statistic: 159.1093217311018 on 1.0 and 1028.0 DF, p-value: -0.0
--------------------------------------------------------------------------------
```

Statsmodels fit report:

```
                         OLS Regression Results
==============================================================================
Dep. Variable: y R-squared: 0.134
Model: OLS Adj. R-squared: 0.133
```

```
Method: Least Squares F-statistic: 159.1
Date: Wed, 02 Sep 2026 Prob (F-statistic): 5.08e-34
Time: 16:40:25 Log-Likelihood: -4287.1
No. Observations: 1030 AIC: 8578.
Df Residuals: 1028 BIC: 8588.
Df Model: 1
Covariance Type: nonrobust
==============================================================================
               coef std err t P>|t| [0.025 0.975]
------------------------------------------------------------------------------
const 29.4668 0.699 42.165 0.000 28.095 30.838
x1 1.0239 0.081 12.614 0.000 0.865 1.183
==============================================================================
Omnibus: 24.083 Durbin-Watson: 1.052
Prob(Omnibus): 0.000 Jarque-Bera (JB): 24.387
Skew: 0.353 Prob(JB): 5.06e-06
Kurtosis: 2.738 Cond. No. 12.5
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

Summary statement: Here the two tools agree on $\hat{y} = 29.4668 + 1.0239\,x$. Each extra kg/m^3 of superplasticizer adds about 1.02 MPa of strength, but this predictor alone only explains 13.4% of the variance ($R^2 = 0.1340$, adj. $R^2 = 0.1332$); the slope is nonetheless highly significant ($F = 159.11$, $p < 0.001$).



(a) Statsmodels

(b) ScalaTion

Figure 6: Compressive strength vs. superplasticizer dosage: actual (black) and predicted (red) values.

2.7 Discussion

Compressive strength depends on cement, water, several admixtures, and curing age all at once, so it is not surprising that no single ingredient explains most of the variance. `cement` is the strongest univariate predictor, but it only reaches $R^2 = 0.25$; `superplasticizer` explains just 13%. Both effects are positive and highly significant, which matches basic concrete chemistry: more cement binder and more superplasticizer, which improves workability at low water content, both tend to raise strength. Still, the scatter in the fit plots is wide. Unlike Auto MPG, this dataset would need a multiple regression across all 8 ingredient and age features to fit well.

3 Airfoil Self-Noise Dataset

3.1 Description

This NASA dataset relates the scaled sound pressure level (decibels) produced by airfoils in a wind tunnel to five aerodynamic and geometric parameters. It contains 1503 rows and 6 columns.

3.2 Preprocessing

The raw data ships as a tab-delimited `.dat` file with no header row. It was converted directly to CSV, adding the column names `frequency`, `angle_of_attack`, `chord_length`, `free_stream_velocity`, `suction_side_displacement_thickness`, and `scaled_sound_pressure_level` (the target) based on the dataset's documentation. There are no missing values and no string columns, so no rows were dropped, and the cleaned shape (1503, 6) matches the documented size exactly. An IQR-rule check flags a substantial number of outliers in `frequency` (86), `angle_of_attack` (30), and `suction_side_displacement_thickness` (124). These reflect the wide range of physical test conditions used in the wind-tunnel experiments (frequency spans 200–20,000 Hz) rather than data-quality problems, so all rows were retained.

3.3 Statistical Summary

Feature	count	mean	std	min	25%	50%	75%	max
frequency	1503.000	2886.381	3152.573	200.000	800.000	1600.000	4000.000	20000.000
angle_of_attack	1503.000	6.782	5.918	0.000	2.000	5.400	9.900	22.200
chord_length	1503.000	0.137	0.094	0.025	0.051	0.102	0.229	0.305
free_stream_velocity	1503.000	50.861	15.573	31.700	39.600	39.600	71.300	71.300
suction_side_displacement_thickness	1503.000	0.011	0.013	0.000	0.003	0.005	0.016	0.058
scaled_sound_pressure_level	1503.000	124.836	6.899	103.380	120.191	125.721	129.995	140.987

3.4 Correlation Analysis

Feature	Correlation with target
frequency	-0.3907
suction_side_displacement_thickness	-0.3127
chord_length	-0.2362
angle_of_attack	-0.1561
free_stream_velocity	0.1251

Table 3 gives the complete feature-by-feature correlation matrix, and Figure 7 shows the corresponding heatmap.

Table 3: Full correlation matrix, Airfoil Self-Noise dataset.

Feature	F	AOA	CL	FSV	SSDT	SSPL
frequency	1.00	-0.27	-0.00	0.13	-0.23	-0.39
angle_of_attack	-0.27	1.00	-0.50	0.06	0.75	-0.16
chord_length	-0.00	-0.50	1.00	0.00	-0.22	-0.24
free_stream_velocity	0.13	0.06	0.00	1.00	-0.00	0.13
suction_side_displacement_thickness	-0.23	0.75	-0.22	-0.00	1.00	-0.31
scaled_sound_pressure_level	-0.39	-0.16	-0.24	0.13	-0.31	1.00

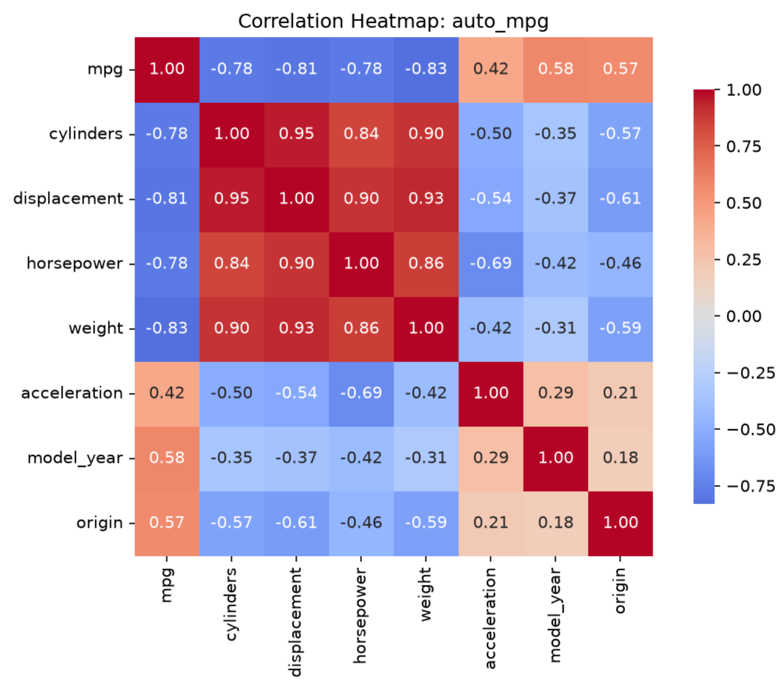


Figure 7: Correlation heatmap for the Airfoil Self-Noise dataset.

3.5 Top Two Predictor Variables

Based on the correlation analysis above, the two predictors most strongly correlated with the scaled sound pressure level are `frequency` ($r = -0.39$) and `suction_side_displacement_thickness` ($r = -0.31$), ahead of `chord_length` ($r = -0.24$) and `angle_of_attack` ($r = -0.16$). These two features are used consistently in both the ScalaTion and Statsmodels simple regressions below.

3.6 Simple Regression Models

Feature: frequency

ScalaTion fit report:

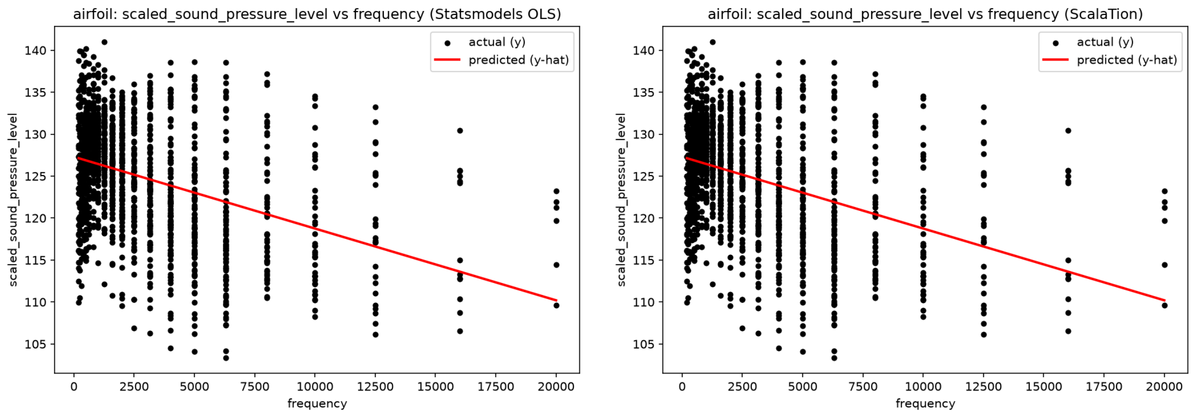
```
REPORT
    ----------------------------------------------------------------------------
    modelName mn = SimpleRegression_frequency
    ----------------------------------------------------------------------------
    hparameter hp = null
    ----------------------------------------------------------------------------
    features fn = Array(one, frequency)
    ----------------------------------------------------------------------------
    parameter b = VectorD(127.304, -0.000854979)
    ----------------------------------------------------------------------------
    fitMap qof = LinkedHashMap(rSq -> 0.152655, rSqBar -> 0.152091, sst -> 71482.377701, sse -> 60570.206223,
         sde -> 6.350305, rse -> 6.352420, mse0 -> 40.299538, rmse -> 6.348192, mae -> 5.068696, smape ->
         4.086777, bmae -> 7.995140, m -> 1503.000000, dfr -> 1.000000, df -> 1501.000000, fStat ->
         270.416272, aic -> -4906.464137, bic -> -4895.833700, mape -> 4.107156, mase -> 2.087582, smapeC ->
         4.089439, picp -> -1.000000, pinc -> -1.000000, ace -> -1.000000, pinaw -> -1.000000, mis ->
         -1.000000, wis -> -1.000000)
    ----------------------------------------------------------------------------

fname = Array(one, frequency)
SUMMARY
```

```
    Parameters/Coefficients:
    Var Estimate Std. Error  t value  Pr(>|t|)  VIF
--------------------------------------------------------------------------------
    x0 127.303738 0.222192 572.944445 0.000000 NA
    x1 -0.000855 0.000052 -16.444339 0.000000 1.000000

    Residual standard error: 6.352420 on 1501.0 degrees of freedom
    Multiple R-squared: 0.152655, Adjusted R-squared: 0.152091
    F-statistic: 270.4162724626191 on 1.0 and 1501.0 DF, p-value: -0.0
--------------------------------------------------------------------------------
```

Statsmodels fit report:

```
                            OLS Regression Results
==============================================================================
Dep. Variable: y R-squared: 0.153
Model: OLS Adj. R-squared: 0.152
Method: Least Squares F-statistic: 270.4
Date: Wed, 02 Sep 2026 Prob (F-statistic): 5.36e-56
Time: 16:40:28 Log-Likelihood: -4910.5
No. Observations: 1503 AIC: 9825.
Df Residuals: 1501 BIC: 9836.
Df Model: 1
Covariance Type: nonrobust
==============================================================================
                coef std err t P>|t| [0.025 0.975]
------------------------------------------------------------------------------
const 127.3037 0.222 572.944 0.000 126.868 127.740
x1 -0.0009 5.2e-05 -16.444 0.000 -0.001 -0.001
==============================================================================
Omnibus: 4.973 Durbin-Watson: 0.255
Prob(Omnibus): 0.083 Jarque-Bera (JB): 5.002
Skew: -0.126 Prob(JB): 0.0820
Kurtosis: 2.873 Cond. No. 5.80e+03
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 5.8e+03. This might indicate that there are
strong multicollinearity or other numerical problems.
```

Summary statement: The two fits again match: $\hat{y} = 127.3037 - 0.0009\,x$. Each additional Hz of frequency lowers sound pressure level by about 0.0009 dB, and this single feature explains only 15.3% of the variance ($R^2 = 0.1527$, adj. $R^2 = 0.1521$), despite a highly significant slope ($F = 270.42$, $p < 0.001$).



(a) Statsmodels (b) ScalaTion

Figure 8: Sound pressure level vs. frequency: actual (black) and predicted (red) values.

Feature: suction_side_displacement_thickness

ScalaTion fit report:

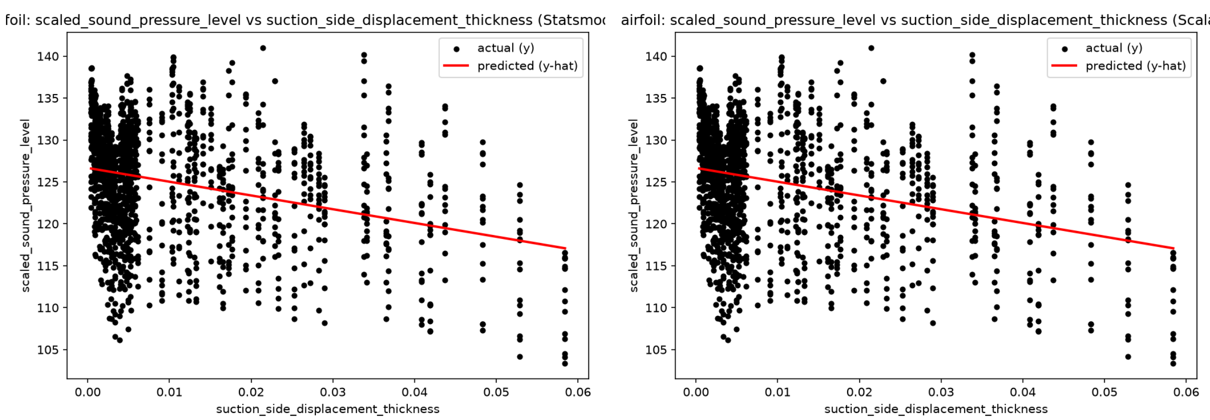
```
REPORT
    ----------------------------------------------------------------------------
    modelName mn = SimpleRegression_suction_side_displacement_thickness
    ----------------------------------------------------------------------------
    hparameter hp = null
    ----------------------------------------------------------------------------
    features fn = Array(one, suction_side_displacement_thickness)
    ----------------------------------------------------------------------------
    parameter b = VectorD(126.663, -164.027)
    ----------------------------------------------------------------------------
    fitMap qof = LinkedHashMap(rSq -> 0.097762, rSqBar -> 0.097161, sst -> 71482.377701, sse -> 64494.101755,
         sde -> 6.552772, rse -> 6.554954, mse0 -> 42.910247, rmse -> 6.550591, mae -> 5.317605, smape ->
         4.284487, bmae -> 8.076464, m -> 1503.000000, dfr -> 1.000000, df -> 1501.000000, fStat ->
         162.641263, aic -> -4953.636279, bic -> -4943.005842, mape -> 4.315273, mase -> 2.108816, smapeC ->
         4.287148, picp -> -1.000000, pinc -> -1.000000, ace -> -1.000000, pinaw -> -1.000000, mis ->
         -1.000000, wis -> -1.000000)
    ----------------------------------------------------------------------------

fname = Array(one, suction_side_displacement_thickness)
SUMMARY
    Parameters/Coefficients:
    Var Estimate Std. Error  t value  Pr(>|t|)  VIF
--------------------------------------------------------------------------------
    x0 126.663189 0.221623 571.526554 0.000000 NA
    x1 -164.027463 12.861784 -12.753088 0.000000 1.000000

    Residual standard error: 6.554954 on 1501.0 degrees of freedom
    Multiple R-squared: 0.097762, Adjusted R-squared: 0.097161
    F-statistic: 162.64126345742537 on 1.0 and 1501.0 DF, p-value: -0.0
--------------------------------------------------------------------------------
```

Statsmodels fit report:

```
                         OLS Regression Results
==============================================================================
Dep. Variable: y R-squared: 0.098
Model: OLS Adj. R-squared: 0.097
Method: Least Squares F-statistic: 162.6
Date: Wed, 02 Sep 2026 Prob (F-statistic): 1.92e-35
Time: 16:40:29 Log-Likelihood: -4957.6
No. Observations: 1503 AIC: 9919.
```

```
Df Residuals: 1501 BIC: 9930.
Df Model: 1
Covariance Type: nonrobust
==============================================================================
               coef std err t P>|t| [0.025 0.975]
------------------------------------------------------------------------------
const 126.6632 0.222 571.527 0.000 126.228 127.098
x1 -164.0275 12.862 -12.753 0.000 -189.256 -138.798
==============================================================================
Omnibus: 26.870 Durbin-Watson: 0.364
Prob(Omnibus): 0.000 Jarque-Bera (JB): 25.468
Skew: -0.280 Prob(JB): 2.95e-06
Kurtosis: 2.694 Cond. No. 76.1
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

Summary statement: ScalaTion and Statsmodels once more agree: $\hat{y} = 126.6632 - 164.0275\,x$. The slope looks dramatic (a full meter of suction-side displacement thickness would cost 164 dB), but this feature only ranges from about 0.0004 to 0.058 m in the data, so realistic changes amount to a few dB at most. The model explains just 9.8% of the variance ($R^2 = 0.0978$, adj. $R^2 = 0.0972$), though the slope remains highly significant ($F = 162.64$, $p < 0.001$).



(a) Statsmodels

(b) ScalaTion

Figure 9: Sound pressure level vs. suction-side displacement thickness: actual (black) and predicted (red) values.

3.7 Discussion

Airfoil Self-Noise has the weakest linear relationships of the three datasets. Even the best single predictor, `frequency`, explains only about 15% of the variance in sound pressure level, and `suction_side_displacement_thickness` explains under 10%. Both effects are negative and statistically significant, but the low R^2 values and wide scatter in the fit plots point to something more complex: aerodynamic self-noise depends on frequency, angle of attack, geometry, and flow velocity together, and turbulent boundary-layer noise does not scale linearly with any one of them. Getting strong predictive accuracy here would likely take a nonlinear or multiple-regression model, or the neural-network approach the original NASA study used.

Cross-Dataset Summary

- **Auto MPG**: strongest simple-regression fit of the three ($R^2 \approx 0.69$ for `weight`); vehicle size/mass is a good linear proxy for fuel economy.
- **Concrete**: moderate fit ($R^2 \approx 0.25$ for `cement`); strength is inherently multivariate.
- **Airfoil**: weakest fit ($R^2 \approx 0.15$ for `frequency`); self-noise generation is a complex, nonlinear aerodynamic phenomenon poorly captured by any single linear predictor.

ScalaTion's `SimpleRegression` and Python's Statsmodels `OLS` produced identical coefficients and R^2 values across all three datasets, since both are solving the same least-squares problem.